

See Problem 39. If

$$\mathbf{w} = -200\mathbf{j}$$

$$\mathbf{F}_1 = (\|\mathbf{F}_1\| \cos 20^\circ)\mathbf{i} + (\|\mathbf{F}_1\| \sin 20^\circ)\mathbf{j}$$

$$\mathbf{F}_2 = -(\|\mathbf{F}_2\| \cos 15^\circ)\mathbf{i} + (\|\mathbf{F}_2\| \sin 15^\circ)\mathbf{j},$$

use (7) to determine the magnitudes of $\mathbf{F}_1$ and $\mathbf{F}_2$. [Hint: Reread (iii) of Definition 7.1.1.]

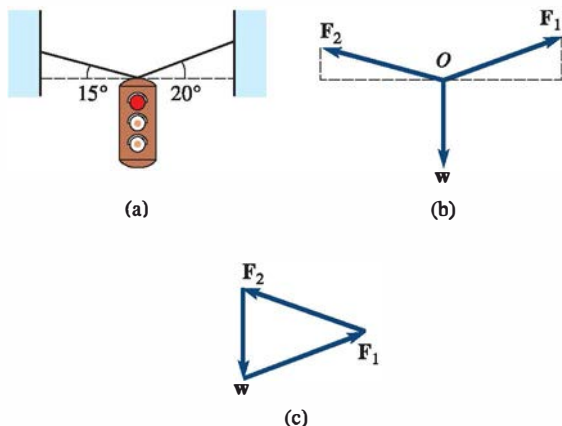


FIGURE 7.1.19 Three force vectors in Problem 46

47. An electric charge Q is uniformly distributed along the y -axis between $y = -a$ and $y = a$. See FIGURE 7.1.20. The total force exerted on the charge q on the x -axis by the charge Q is $\mathbf{F} = F_x\mathbf{i} + F_y\mathbf{j}$, where

$$F_x = \frac{qQ}{4\pi\epsilon_0} \int_{-a}^a \frac{L dy}{2a(L^2 + y^2)^{3/2}}$$

and

$$F_y = -\frac{qQ}{4\pi\epsilon_0} \int_{-a}^a \frac{y dy}{2a(L^2 + y^2)^{3/2}}.$$

Determine $\mathbf{F}$.

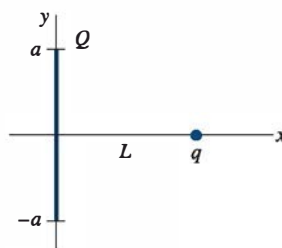


FIGURE 7.1.20 Charge on x -axis in Problem 47

48. Using vectors, show that the diagonals of a parallelogram bisect each other. [Hint: Let M be the midpoint of one diagonal and N the midpoint of the other.]
49. Using vectors, show that the line segment between the midpoints of two sides of a triangle is parallel to the third side and half as long.
50. An airplane starts from an airport located at the origin O and flies 150 miles in the direction 20° north of east to city A . From A , the airplane then flies 200 miles in the direction 23° west of north to city B . From B , the airplane flies 240 miles in the direction 10° south of west to city C . Express the location of city C as a vector $\mathbf{r}$ as shown in FIGURE 7.1.21. Find the distance from O to C .

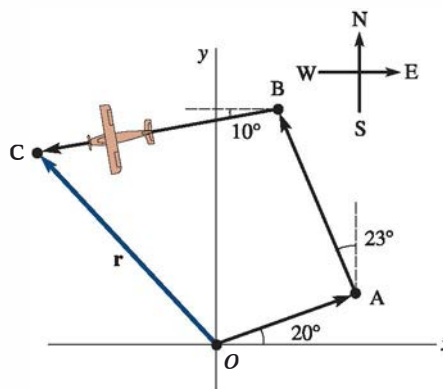


FIGURE 7.1.21 Airplane in Problem 50

7.2 Vectors in 3-Space

Introduction In the plane, or 2-space, one way of describing the position of a point P is to assign to it coordinates relative to two mutually orthogonal, or perpendicular, axes called the x - and y -axes. If P is the point of intersection of the line $x = a$ (perpendicular to the x -axis) and the line $y = b$ (perpendicular to the y -axis), then the **ordered pair** (a, b) is said to be the **rectangular** or **Cartesian coordinates** of the point. See FIGURE 7.2.1. In this section we extend the notions of Cartesian coordinates and vectors to three dimensions.

Rectangular Coordinate System in 3-Space In three dimensions, or **3-space**, a rectangular coordinate system is constructed using three mutually orthogonal axes. The point at which these axes intersect is called the origin O . These axes, shown in FIGURE 7.2.2(a), are labeled in accordance with the so-called **right-hand rule**: If the fingers of the right hand, pointing in the direction of the positive x -axis, are curled toward the positive y -axis, then the thumb will point in the direction of a new axis perpendicular to the plane of the x - and y -axes. This new axis is labeled the z -axis.

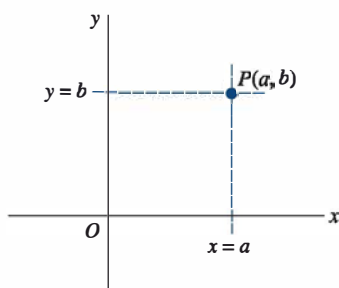


FIGURE 7.2.1 Rectangular coordinates in 2-space

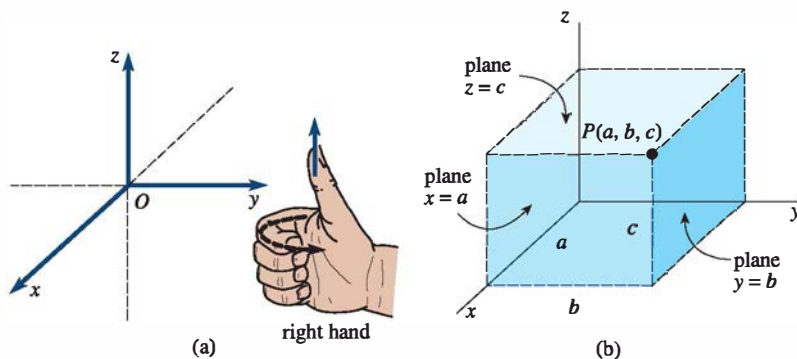


FIGURE 7.2.2 Rectangular coordinates in 3-space

The dashed lines in Figure 7.2.2(a) represent the negative axes. Now, if

$$x = a, \quad y = b, \quad z = c$$

are planes perpendicular to the x -axis, y -axis, and z -axis, respectively, then the point P at which these planes intersect can be represented by an **ordered triple** of numbers (a, b, c) said to be the **rectangular** or **Cartesian coordinates** of the point. The numbers a , b , and c are, in turn, called the **x -**, **y -**, and **z -coordinates** of $P(a, b, c)$. See Figure 7.2.2(b).

Octants Each pair of coordinate axes determines a **coordinate plane**. As shown in FIGURE 7.2.3, the x - and y -axes determine the xy -plane, the x - and z -axes determine the xz -plane, and so on. The coordinate planes divide 3-space into eight parts known as **octants**. The octant in which all three coordinates of a point are *positive* is called the **first octant**. There is no agreement for naming the other seven octants.

The following table summarizes the coordinates of a point either on a coordinate axis or in a coordinate plane. As seen in the table, we can also describe, say, the xy -plane by the simple equation $z = 0$. Similarly, the xz -plane is $y = 0$ and the yz -plane is $x = 0$.

Axes	Coordinates	Plane	Coordinates
x	$(a, 0, 0)$	xy	$(a, b, 0)$
y	$(0, b, 0)$	xz	$(a, 0, c)$
z	$(0, 0, c)$	yz	$(0, b, c)$

EXAMPLE 1 Graphs of Three Points

Graph the points $(4, 5, 6)$, $(3, -3, -1)$, and $(-2, -2, 0)$.

SOLUTION Of the three points shown in FIGURE 7.2.4, only $(4, 5, 6)$ is in the first octant. The point $(-2, -2, 0)$ is in the xy -plane.

Distance Formula To find the **distance** between two points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ in 3-space, let us first consider their projection onto the xy -plane. As seen in FIGURE 7.2.5, the distance between $(x_1, y_1, 0)$ and $(x_2, y_2, 0)$ follows from the usual distance formula in the plane and is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. If the coordinates of P_3 are (x_2, y_2, z_1) , then the Pythagorean theorem applied to the right triangle $P_1P_2P_3$ yields

$$[d(P_1, P_2)]^2 = [\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}]^2 + |z_2 - z_1|^2$$

$$\text{or} \quad d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}. \quad (1)$$

EXAMPLE 2 Distance Between Two Points

Find the distance between $(2, -3, 6)$ and $(-1, -7, 4)$.

SOLUTION Choosing P_2 as $(2, -3, 6)$ and P_1 as $(-1, -7, 4)$, formula (1) gives

$$d = \sqrt{(2 - (-1))^2 + (-3 - (-7))^2 + (6 - 4)^2} = \sqrt{29}. \quad \equiv$$

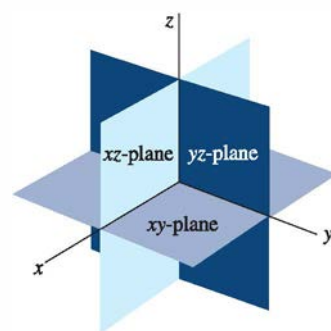


FIGURE 7.2.3 Octants

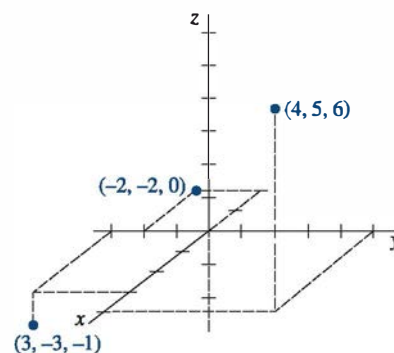


FIGURE 7.2.4 Points in Example 1

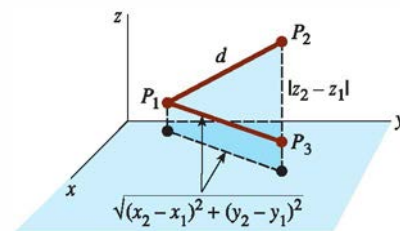


FIGURE 7.2.5 Distance d between two points in 3-space

Midpoint Formula The formula for finding the midpoint of a line segment between two points in 2-space carries over in an analogous fashion to 3-space. If $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ are two distinct points, then the **coordinates of the midpoint** of the line segment between them are

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right). \quad (2)$$

EXAMPLE 3 Coordinates of a Midpoint

Find the coordinates of the midpoint of the line segment between the two points in Example 2.

SOLUTION From (2) we obtain

$$\left(\frac{2 + (-1)}{2}, \frac{-3 + (-7)}{2}, \frac{6 + 4}{2} \right) \quad \text{or} \quad \left(\frac{1}{2}, -5, 5 \right). \quad \equiv$$

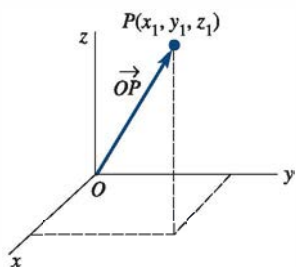


FIGURE 7.2.6 Position vector

Vectors in 3-Space A vector **a** in 3-space is any ordered triple of real numbers

$$\mathbf{a} = \langle a_1, a_2, a_3 \rangle,$$

where a_1 , a_2 , and a_3 are the **components** of the vector. The set of all vectors in 3-space will be denoted by the symbol R^3 . The **position vector** of a point $P(x_1, y_1, z_1)$ in space is the vector $\vec{OP} = \langle x_1, y_1, z_1 \rangle$ whose initial point is the origin O and whose terminal point is P . See FIGURE 7.2.6.

The component definitions of addition, subtraction, scalar multiplication, and so on are natural generalizations of those given for vectors in R^2 . Moreover, the vectors in R^3 possess all the properties listed in Theorem 7.1.1.

Definition 7.2.1 Component Definitions in 3-Space

Let $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$ be vectors in R^3 .

- (i) Addition: $\mathbf{a} + \mathbf{b} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$
- (ii) Scalar multiplication: $k\mathbf{a} = \langle ka_1, ka_2, ka_3 \rangle$
- (iii) Equality: $\mathbf{a} = \mathbf{b}$ if and only if $a_1 = b_1, a_2 = b_2, a_3 = b_3$
- (iv) Negative: $-\mathbf{b} = (-1)\mathbf{b} = \langle -b_1, -b_2, -b_3 \rangle$
- (v) Subtraction: $\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b}) = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$
- (vi) Zero vector: $\mathbf{0} = \langle 0, 0, 0 \rangle$
- (vii) Magnitude: $\|\mathbf{a}\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$

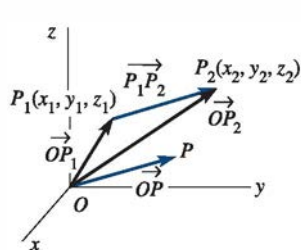


FIGURE 7.2.7 $\vec{OP}$ and $\vec{P_1P_2}$ are the same vector

If $\vec{OP_1}$ and $\vec{OP_2}$ are the position vectors of the points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$, then the vector $\vec{P_1P_2}$ is given by

$$\vec{P_1P_2} = \vec{OP_2} - \vec{OP_1} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle. \quad (3)$$

As in 2-space, $\vec{P_1P_2}$ can be drawn either as a vector whose initial point is P_1 and whose terminal point is P_2 or as position vector $\vec{OP}$ with terminal point

$$P(x_2 - x_1, y_2 - y_1, z_2 - z_1).$$

See FIGURE 7.2.7.

EXAMPLE 4 Vector Between Two Points

Find the vector $\vec{P_1P_2}$ if the points P_1 and P_2 are given by $P_1(4, 6, -2)$ and $P_2(1, 8, 3)$, respectively.

SOLUTION If the position vectors of the points are $\vec{OP_1} = \langle 4, 6, -2 \rangle$ and $\vec{OP_2} = \langle 1, 8, 3 \rangle$, then from (3) we have

$$\vec{P_1P_2} = \vec{OP_2} - \vec{OP_1} = \langle 1 - 4, 8 - 6, 3 - (-2) \rangle = \langle -3, 2, 5 \rangle. \quad \equiv$$

EXAMPLE 5 A Unit Vector

Find a unit vector in the direction of $\mathbf{a} = \langle -2, 3, 6 \rangle$.

SOLUTION Since a unit vector has length 1, we first find the magnitude of $\mathbf{a}$ and then use the fact that $\mathbf{a}/\|\mathbf{a}\|$ is a unit vector in the direction of $\mathbf{a}$. The magnitude of $\mathbf{a}$ is

$$\|\mathbf{a}\| = \sqrt{(-2)^2 + 3^2 + 6^2} = \sqrt{49} = 7.$$

A unit vector in the direction of $\mathbf{a}$ is

$$\frac{\mathbf{a}}{\|\mathbf{a}\|} = \frac{1}{7} \langle -2, 3, 6 \rangle = \left\langle -\frac{2}{7}, \frac{3}{7}, \frac{6}{7} \right\rangle \quad \equiv$$

The $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ Vectors We saw in the preceding section that the set of two unit vectors $\mathbf{i} = \langle 1, 0 \rangle$ and $\mathbf{j} = \langle 0, 1 \rangle$ constitute a basis for the system of two-dimensional vectors. That is, any vector $\mathbf{a}$ in 2-space can be written as a linear combination of $\mathbf{i}$ and $\mathbf{j}$: $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j}$. Likewise any vector $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ in 3-space can be expressed as a linear combination of the unit vectors

$$\mathbf{i} = \langle 1, 0, 0 \rangle, \quad \mathbf{j} = \langle 0, 1, 0 \rangle, \quad \mathbf{k} = \langle 0, 0, 1 \rangle.$$

To see this we use (i) and (ii) of Definition 7.2.1 to write

$$\begin{aligned} \langle a_1, a_2, a_3 \rangle &= \langle a_1, 0, 0 \rangle + \langle 0, a_2, 0 \rangle + \langle 0, 0, a_3 \rangle \\ &= a_1 \langle 1, 0, 0 \rangle + a_2 \langle 0, 1, 0 \rangle + a_3 \langle 0, 0, 1 \rangle; \end{aligned}$$

that is,

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}.$$

The vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ illustrated in **FIGURE 7.2.8(a)** are called the **standard basis** for the system of three-dimensional vectors. In Figure 7.2.8(b) we see that a position vector $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ is the sum of the vectors $a_1\mathbf{i}$, $a_2\mathbf{j}$, and $a_3\mathbf{k}$, which lie along the coordinate axes and have the origin as a common initial point.

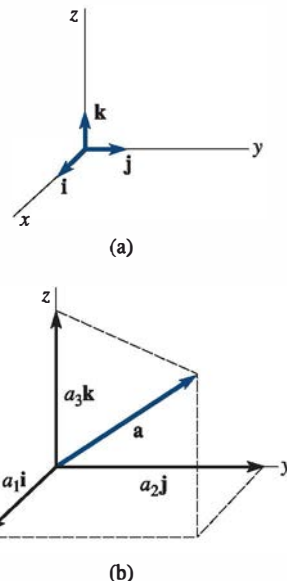


FIGURE 7.2.8 $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ form a basis for $\mathbb{R}^3$

EXAMPLE 6 Using the $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ Vectors

The vector $\mathbf{a} = \langle 7, -5, 13 \rangle$ is the same as $\mathbf{a} = 7\mathbf{i} - 5\mathbf{j} + 13\mathbf{k}$. $\equiv$

When the third dimension is taken into consideration, any vector in the xy -plane is equivalently described as a three-dimensional vector that lies in the coordinate plane $z = 0$. Although the vectors $\langle a_1, a_2 \rangle$ and $\langle a_1, a_2, 0 \rangle$ are technically not equal, we shall ignore the distinction. That is why, for example, we denoted $\langle 1, 0 \rangle$ and $\langle 1, 0, 0 \rangle$ by the same symbol $\mathbf{i}$. But to avoid any possible confusion, hereafter we shall always consider a vector a three-dimensional vector, and the symbols $\mathbf{i}$ and $\mathbf{j}$ will represent only $\langle 1, 0, 0 \rangle$ and $\langle 0, 1, 0 \rangle$, respectively. Similarly, a vector in either the xy -plane or the xz -plane must have one zero component. In the yz -plane, a vector

$$\mathbf{b} = \langle 0, b_2, b_3 \rangle \quad \text{is written} \quad \mathbf{b} = b_2\mathbf{j} + b_3\mathbf{k}.$$

In the xz -plane, a vector

$$\mathbf{c} = \langle c_1, 0, c_3 \rangle \quad \text{is the same as} \quad \mathbf{c} = c_1\mathbf{i} + c_3\mathbf{k}.$$

EXAMPLE 7 Vector in xz -Plane

(a) The vector $\mathbf{a} = 5\mathbf{i} + 3\mathbf{k} = 5\mathbf{i} + 0\mathbf{j} + 3\mathbf{k}$ lies in the xz -plane and can also be written as $\mathbf{a} = \langle 5, 0, 3 \rangle$.

(b) $\|5\mathbf{i} + 3\mathbf{k}\| = \sqrt{5^2 + 0^2 + 3^2} = \sqrt{25 + 9} = \sqrt{34}$ $\equiv$

EXAMPLE 8 Linear Combination

If $\mathbf{a} = 3\mathbf{i} - 4\mathbf{j} + 8\mathbf{k}$ and $\mathbf{b} = \mathbf{i} - 4\mathbf{k}$, find $5\mathbf{a} - 2\mathbf{b}$.

SOLUTION We treat $\mathbf{b}$ as a three-dimensional vector and write, for emphasis, $\mathbf{b} = \mathbf{i} + 0\mathbf{j} - 4\mathbf{k}$. From

$$5\mathbf{a} = 15\mathbf{i} - 20\mathbf{j} + 40\mathbf{k} \quad \text{and} \quad 2\mathbf{b} = 2\mathbf{i} + 0\mathbf{j} - 8\mathbf{k}$$

we get

$$\begin{aligned} 5\mathbf{a} - 2\mathbf{b} &= (15\mathbf{i} - 20\mathbf{j} + 40\mathbf{k}) - (2\mathbf{i} + 0\mathbf{j} - 8\mathbf{k}) \\ &= 13\mathbf{i} - 20\mathbf{j} + 48\mathbf{k}. \end{aligned} \quad \equiv$$